

The impact of consistent query and target structures in embedding model evaluation

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Structure Retention in Embedding Spaces as a Predictor of Benchmark Performance

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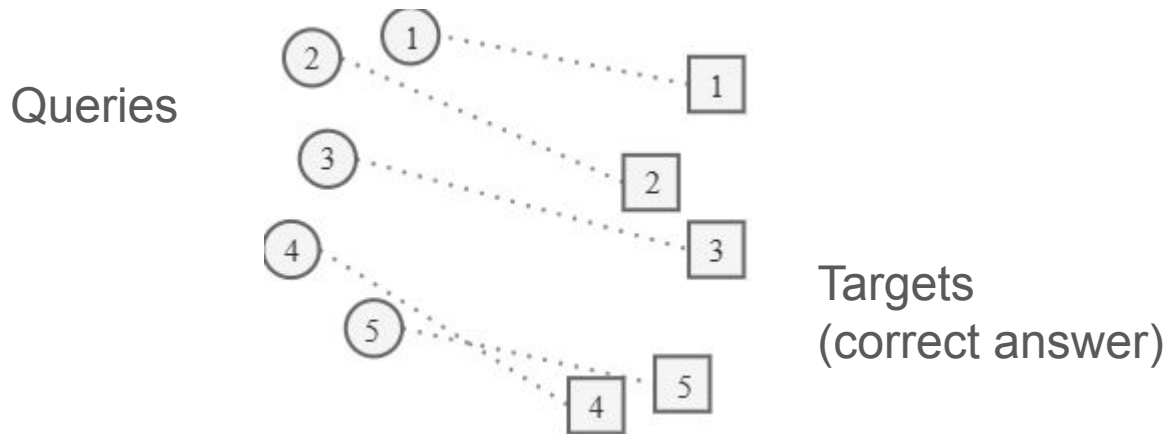
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Summary of methods

- Nearest Neighbor overlap
- Analysis of magnitude differences in ICA components



Summary of methods

3.4 Neighborhood Retention

We evaluate the neighborhood retention between the query and target (t, t') as

$$\text{Ret}_k = \sum_{(t, t') \in D} \frac{\text{Jacc}(k\text{NN}(f(t)), k\text{NN}(f(t')))}{k}$$

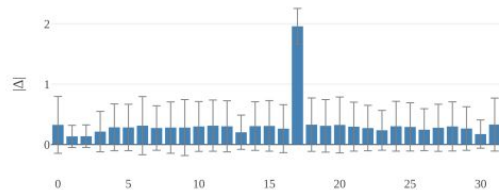
where $k\text{NN}(f(t))$ is the set of k nearest neighbors for the embedding of text t and $\text{Jacc}(\cdot, \cdot)$ is the Jaccard-overlap.

Summary of methods

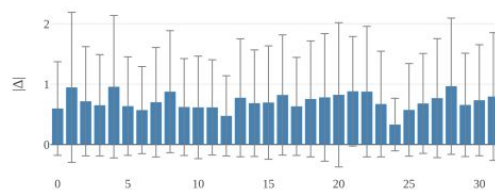
Our motivation is as follows: if modern embedding models exhibit structures described above, then a simple linear method such as ICA should be able to isolate a dominant concept-level signal (e.g., language) in their embedding spaces. To illustrate the intuition, consider a dataset of N translation pairs $D = [(t_1, t'_1), (t_2, t'_2), \dots, (t_N, t'_N)]$ from English texts t_i (queries) to French t'_i (targets). Let $f(t) \in \mathbb{R}^d$ denote the embedding of t . We fit ICA on the embeddings of both sides of the pairs,

$$\text{ICA} \stackrel{\text{fit}}{\leftarrow} \{f(t_1), \dots, f(t_N), f(t'_1), \dots, f(t'_N)\}.$$

If language is, as assumed, the primary systematic difference between queries and targets and is represented approximately linearly, ICA should allocate a dedicated component to this signal.



(a) Intfloat/multilingual-e5-large-instruct



(b) Google/embedding-gemma-300m

Figure 1: Average absolute difference ($|\Delta|$) over English-French translation pairs on ICA (dim=32) transformed embeddings per output dimension, with standard deviations as error bars. Multilingual-e5-large-instruct, which receives high scores for English-French task, shows a characteristic “peak”, while embedding-gemma-300 does not, and correspondingly receives a lower score for the same task.

Results

Table 3: Spearman correlation ($\uparrow$) for neighborhood retention with MTEB scores, with bolded values for strong relationship (≥ 0.7). Star (*) indicates significance at level $p < 0.05$, dagger ($\dagger$) at $p < 0.01$, and double dagger ($\ddagger$) at $p < 0.001$.

Dataset	Unprompted	Prompted
ARCChallenge	0.890$\ddagger$	0.945$\ddagger$
RTE3:eng	-0.230	-0.058
SummEval	0.643 $\ddagger$	0.588 $\dagger$
WebFAQ:bul	0.926$\ddagger$	0.866$\ddagger$
Tatoeba:fin-eng	0.968$\ddagger$	0.959$\ddagger$

Results

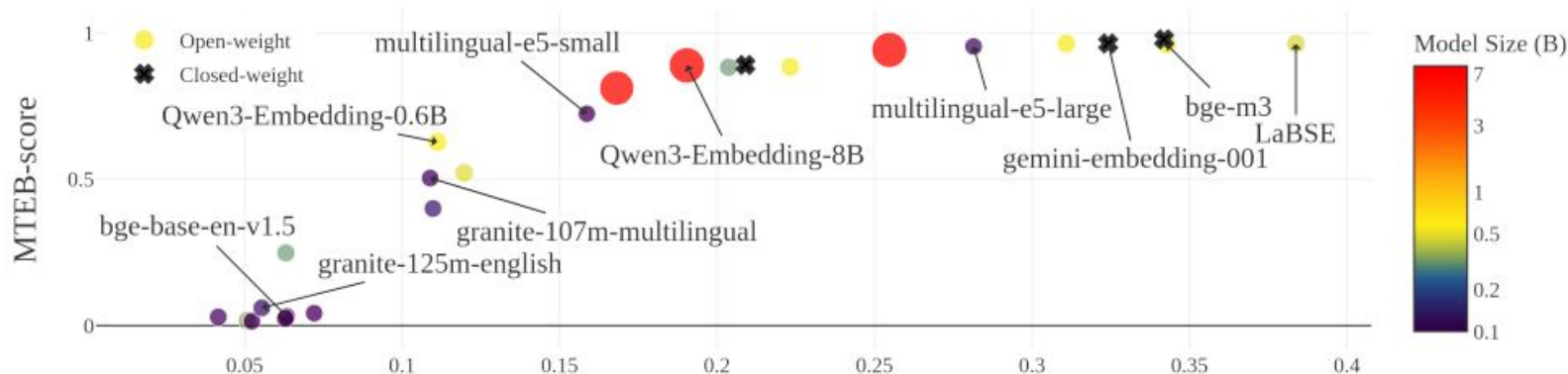
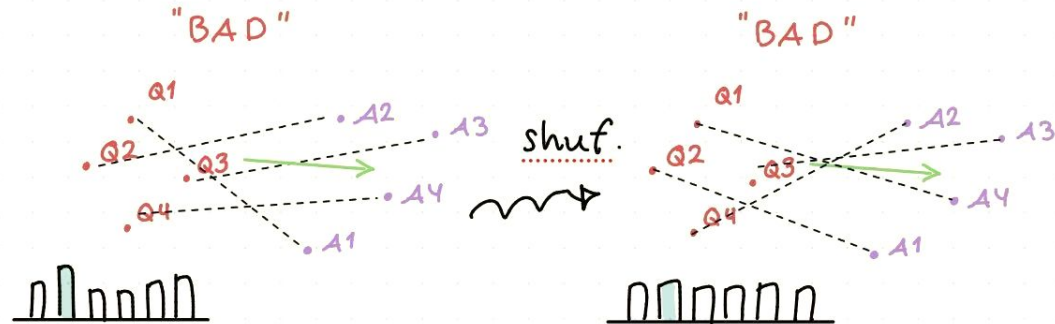
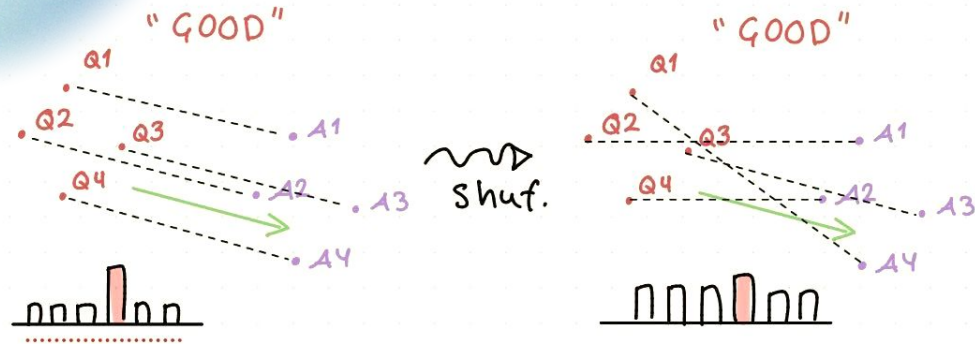


Figure 2: Visualization of the relationship between Ret_k (horizontal) and the MTEB-score for Tatoeba:fin-eng task, which shows one of the strongest correlations in Tatoeba (**0.968[†]**). Annotated model names show that multilingual models outperform their English-only counterparts, and that model size is not the only contributing factor.

Neighborhood and ICA



**Thank you for
listening!**

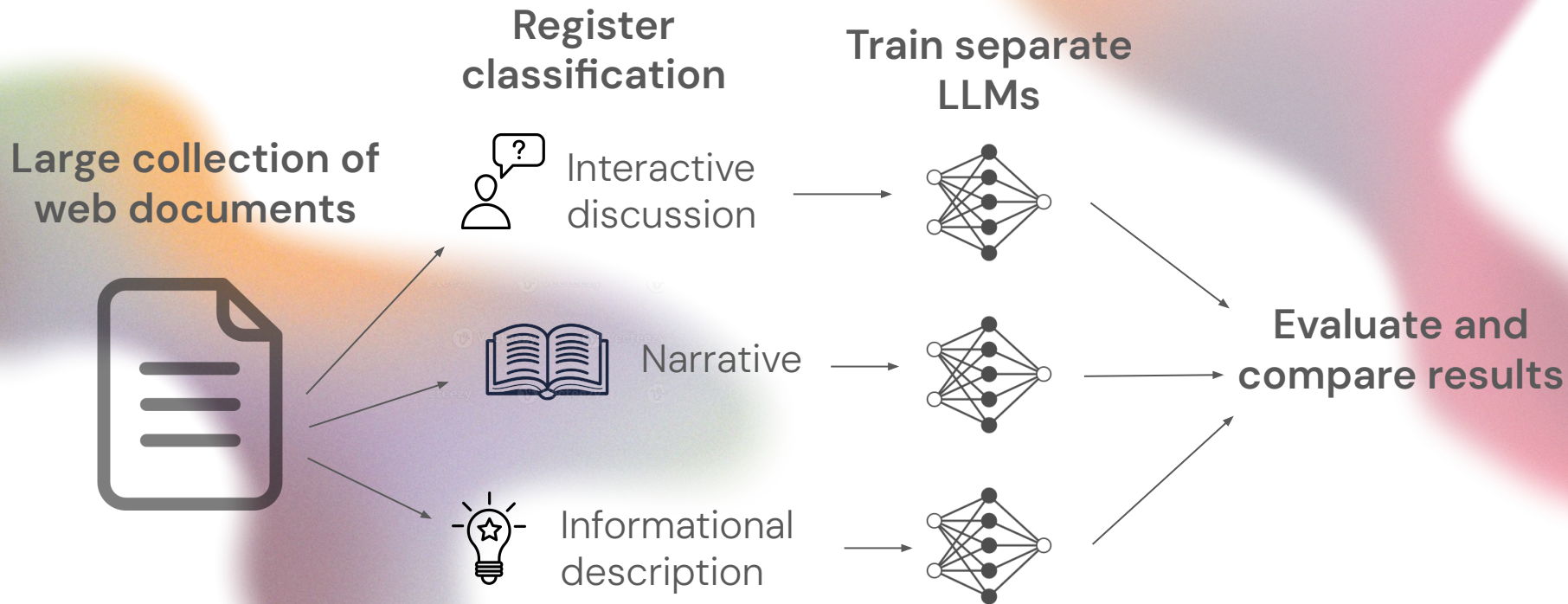


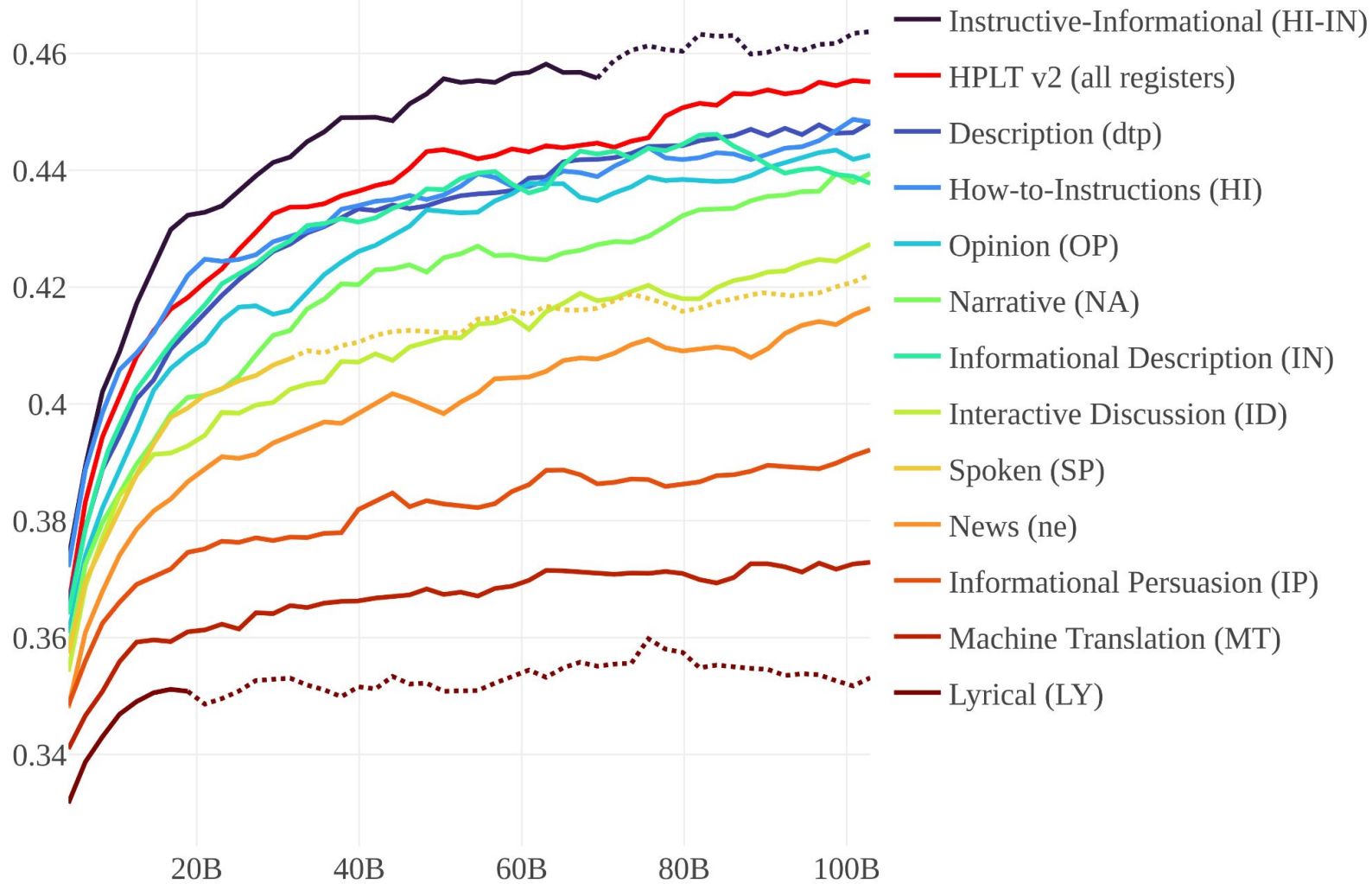
The impact of registers on LLM performance

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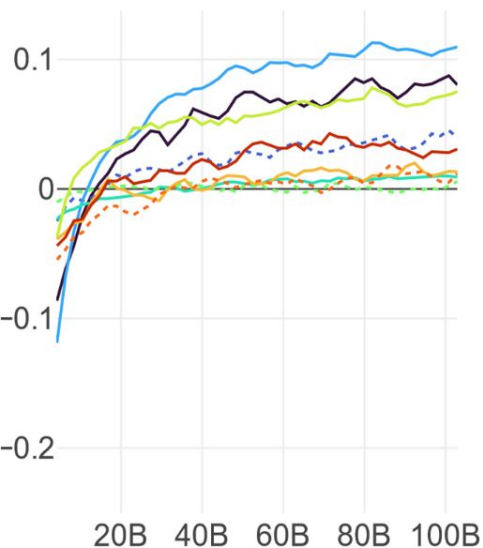
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Summary of method

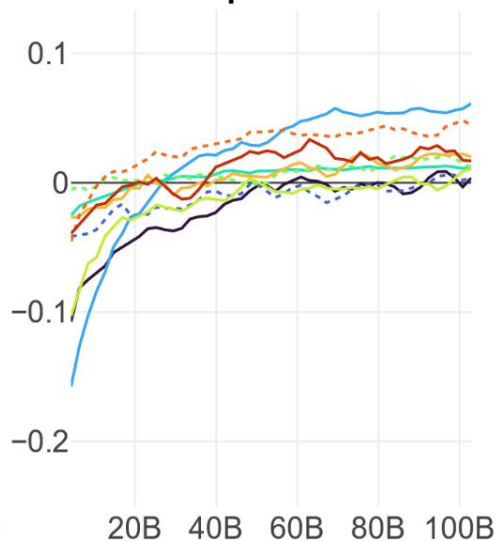




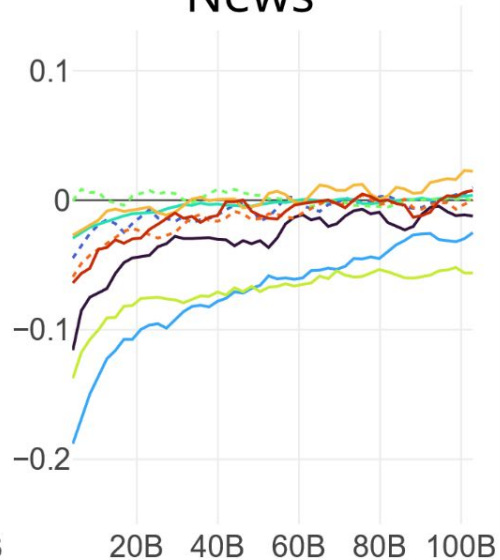
Instructive-Inform.



Opinion



News



Evaluation sets

- ARC Easy
- - - ARC Challenge
- Hellaswag
- MMLU
- - - SIQA
- PIQA
- WinoGrande
- - - CommonsenseQA
- OpenbookQA

**Thank you for
listening!**

